

Viren Kumar

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OBJECTIVE

Undergraduate in Computer Engineering and Applied Mathematics seeking higher level work with Markov Chains, Mathematical Modeling, and Systems Optimization in applied problems.

EDUCATION

California State Polytechnic University, Pomona

August 2024 – May 2027 expected

Dual Major: Bachelor of Science, Computer Engineering
Bachelor of Science, Applied Mathematics

RESEARCH AND TECHNICAL EXPERIENCE

McNair Scholar and Undergraduate Researcher

August 2025 -- Present

McNair Scholars Program | Cal Poly Pomona

Pomona, CA

- Conducted research under Dr. Aly on post-quantum encryption for CubeSat communication as part of the NASA ORBIT competition.
- Focused on structural and mathematical validation of Applying ASCON to the Kyber protocols, including using group theory to verify monoid behavior and lattice-like structure.
- Phase III contestant with team presenting work at Johnson Space Center in Houston.

CubeSTEP Flight Software Lead

December 2025 -- Present

CubeSTEP | Cal Poly Pomona

Pomona, CA

- Collaborate with the JPL F' team to develop flight-ready software for an incoming satellite mission.
- Co-author of paper and presented poster on "Mission-Oriented Flight Software Development: CubeSTEP," for SMC-IT 2026.
- Promoted to software lead due to exceptional performance on August 2026

Research Volunteer

June 2025 -- August 2025

NCUHS | UCLA

Los Angeles, CA

- Collaborated with Dr. Guido Faas on fabrication of micro-sensors and chips for implantation in mice for pain and migraine research.
- Supported chip handling, microfabrication preparation, and technical lab workflows in a biomedical research setting.

Founding Member and Thermal Engineering Lead

August 2024 -- December 2025

CPP Smartphone Project | Cal Poly Pomona

Pomona, CA

- Helped establish project scope and secure funding for a student-led effort to design a smartphone system using FPGA prototyping.
- Led a five-person thermal engineering team focused on body design, thermal management, and sustained performance.
- Coordinated with hardware, system-on-chip, and systems teams to manage thermal and energy budgets; used FreeCAD and Ansys for design iteration.

CPP Problem Solving Team Member

Spring 2024 -- Present

Problem Solving Team | Cal Poly Pomona

Pomona, CA

- Solve problems from mathematical journals and prepared formal solutions in LaTeX for submission.
- Personally submitted verified solutions to CRUX, PME, and CMJ.
- Became main point of contact for proof-writing and inequality problem solving.

Materials Engineering Intern

June 2022 -- August 2022

LyTen Inc.

San Jose, CA

- worked on creating and testing various composites impregnated with different variations of Graphene nanoplateles
- Through labwork formalized and presented a method to filter for grain size with a 70% higher yeild, data was organized and calculated through matlab and excel and presented via powerpoint.

PUBLICATIONS AND MANUSCRIPTS

- Mission-Oriented Flight Software Development: CubeSTEP** August 5, 2026
- 2nd Co-author; publication for SMC-IT 2026.
- From TinyML Model to Open Silicon, A CFU Playground Pipeline for AI design on GF180MCU** August 12, 2026
- 1st Author; publication for STARS 2026.
- Ascon-Kyber for Quantum-Resilient Space Telemetry** May 22, 2026
- 1st author; publication for McNair Symposium 2026.

PRESENTATIONS, PANELS, AND SELECTED TALKS

- Various Steady State Properties of Markov Chains and Processes** Nov 5 2026
- Set to present at research and progress on various steady state properties of Markov Chains, at MAA Southern California-Nevada Section Fall Meeting, 2026 AMS Fall Western Sectional Meeting, and JMM 2027.
- MDO to Software Design** August 5 2026
- Presented poster at SMC-IT with same name as submitted paper. Focused on methodology for turning MDO(Mission Design Optimizations) into actionable software procedures.
- Markov Processes and Weak Stationarity** April 15, 2026
- Presented a lemma showing that a Markov process at steady state is weakly stationary. At Cal Poly Pomona's 2026 Science Symposium.
- Quaternions: An introduction of Abstract Structures** April 8, 2026
- Presented a poster on Cayley-Dickson constructions at SIAM 2026 Symposium at Cal Poly Pomona
- AI Subject Matter Expert Panelist** June 4, 2025
- Panelist as an AI subject matter expert for the OAI Faculty and Staff Summer Conference for the Cal Poly Pomona Student Perspectives on the AI-Driven Future of Education and Work for STEM to Social Science Fields event.
- Thermal Design Optimization for Mobile Devices** August 7, 2025
- Presented research proposal at CARS Conference, Cal Poly Pomona.
- Negative Probabilities in Markov Chains** October 27, 2024
- Presented Dr. Krinik's research at the Fall Western Sectional Meeting, University of California, Riverside.

OTHER CONFERENCES AND WORKSHOPS

- Forum for Diversity in Graduate Education, University of California, SCSU, October 3, 2026
- 12th International Conference on Space Mission Challenges for Information Technology (SMC-IT), August 3–6 2026
- 5th Annual Quantum Engineering Workshop, University of Southern California, May 29, 2025
- IPAM PUMA: Practicum for Undergraduate Mathematicians in Topology, UCLA, April 12–13, 2025
- Forum for Diversity in Graduate Education, University of California, Riverside, October 19, 2024
- 4th Annual Quantum Engineering Workshop, University of Southern California, May 30, 2024

SELECTED COURSEWORK

Mathematics and Statistics

- MAT 5300: Random Processes
- MAT 4010: Numerical Analysis I
- MAT 4190: Advanced Linear Algebra
- MAT 4170: Abstract Algebra I
- MAT 3470: Combinatorics
- MAT 3140: Real Analysis I
- MAT 4990: Introduction to Mathematical Research with negative states in Markov chains

Electrical and Computer Engineering

- ECE 4300: Computer Architecture
- ECE 3300/L: Digital Circuit Design with Verilog
- ECE 2200/L: Microelectronic Circuits
- ECE 3101/L: Signals and Systems
- ECE 2300/L: Digital Logic Design
- ECE 2101/L: Electrical Circuit Analysis II
- ECE 4990: Engineering Research: Computer Vision
- ECE 4990: Engineering Research: Wireless and Mobile Systems

MEMBERSHIPS, AFFILIATIONS, AND SERVICE

- McNair Scholars Program
- Engineering BRIDGE
- SIAM Vice President and Communications Liaison
- CSU-SPARA
- RAMP: Reading, Advising, and Mentoring Program
- MEP-WISE: Maximizing Engineering Potential
- CPP branch IEEE Secretary